

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2219

Monfort 230 kV
March 25, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2219. The Interconnection Customer proposes a 162.5 MW wind facility interconnecting at the Monfort 230 kV substation in the Indiana area. The request was received on October 31, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 622 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,473,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Lonsdale-Ne Harriman 100 KV L	Wabash-Monfort 230 KV	105.3	128	8.9
Timber Switch-W Melrose 230 KV	Trego-Swan Vly 500 KV Line	100.1	929	5.7

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ne Harriman-Lonsdale 100 KV line	Reconductor line	\$6,168,000	25% (\$1,555,000)
Timber Switch-W Melrose 230 KV	Reconductor line	\$43,626,000	2% (\$918,000)
	Total allocated to project		\$2,473,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.